

# Writing the methods section

Project: Reproducible AI Research Study

## 1. Prepare the purpose, prerequisites and materials

### Teach the learning cycle

Primary target: Explain the mechanism and terminology of writing the methods section

#### Session · Sequence and minutes · Resources

**Session:** Session 1 · Understand · 45 min

**Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

**Resources:** Sections 1–6; record the brief recall in the workbook.

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**Session:** Session 2 · Practise · 45 min

**Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

**Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

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**Session:** Session 3 · Investigate and transfer · 45 min

**Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

**Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

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This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

## Retrieval answer guidance

### Figures that preserve evidence

Without opening the old lesson, explain one key idea from Figures that preserve evidence. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the mechanism and terminology of figures that preserve evidence

Reference example, not the only acceptable wording: Teacher explanation: Figures should preserve denominators, scales and uncertainty rather than exaggerate the preferred comparison. A plot is a representation of evidence, not extra evidence. Label axes, units, cases and scope. For simple comparative bars use a zero baseline or explicitly justify another plot. Show condition breakdowns and unassessed counts where relevant.

### Robustness across conditions

Without opening the old lesson, explain one key idea from Robustness across conditions. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the mechanism and terminology of robustness across conditions

Reference example, not the only acceptable wording: Teacher explanation: Robustness checks examine a frozen procedure under declared varied conditions. Performance differences do not by themselves identify cause. Keep baseline, metric and eligibility aligned across conditions. Report condition counts and failures separately; new conditions may extend evidence but cannot guarantee all conditions.

### Reproducible environment and records

Without opening the old lesson, explain one key idea from Reproducible environment and records. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the mechanism and terminology of reproducible environment and records

Reference example, not the only acceptable wording: Teacher explanation: A reproducible environment record links code, data, configuration and runtime dependencies to results. A seed alone is insufficient. Record exact data IDs/splits, code version, dependency versions, parameters, randomness policy and raw outputs. Verify hashes for integrity without treating them as correctness proofs.

Two 45-minute sessions. A methods section should let a reader reconstruct the design and computation. Detail should explain choices that affect evidence. Use the supplied constructed scenarios to make the mechanism inspectable. Collect W4 before sharing solutions. Students should name intermediate evidence and limitations; classroom pacing and learning effectiveness remain unvalidated.

### Outcomes

Explain the mechanism and terminology of writing the methods section

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

## Preparation

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

## Prerequisite answer

A result depends on the inputs, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from predictions or unmeasured conditions. If the prior concept cannot be explained, use a short retrieval activity before this chapter.

## End goal for the pictorial case

Explain writing the methods section using the records and boundaries of this fictional investigation.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

## 2. Explain and model the mechanism

A methods section should let a reader reconstruct the design and computation. Detail should explain choices that affect evidence.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing. These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Report question, units, sampling, permissions, data roles, baseline, procedure, metrics, uncertainty assumptions, missing/exclusion handling and versions. Distinguish prespecified and later changes.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same inputs. If a required input is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

### Term · Meaning

**Term:** Methods section

**Meaning:** The record of study design and computation

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**Term:** Design choice

**Meaning:** A decision affecting what evidence means

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**Term:** Analysis assumption

**Meaning:** A condition required by a statistical procedure

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## Writing the methods section — evidence trace

### Input and scope · Check or calculation · Conclusion and limit

**Input and scope:** Methods says used AI on data and got good scores; code and split are absent.

**Check or calculation:** Report question, units, sampling, permissions, data roles, baseline, procedure, metrics, uncertainty assumptions, missing/exclusion handling and versions. Distinguish prespecified and later changes.

**Conclusion and limit:** It omits model/procedure version, data identity, units/splits, baseline, metric and scoring. Reproduction and validity cannot be assessed from the description.

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The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: It omits model/procedure version, data identity, units/splits, baseline, metric and scoring. Reproduction and validity cannot be assessed from the description. Qualification: A detailed methods section can reveal limitations; it does not guarantee a correct design. Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the mechanism; reject a claim resting on unmeasured evidence.

## Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Methods says used AI on data and got good scores; code and split are absent.

It omits model/procedure version, data identity, units/splits, baseline, metric and scoring. Reproduction and validity cannot be assessed from the description.

Compare the intermediate evidence with the final statement. Explain why each number or decision follows from the stated inputs. The conclusion belongs to this scenario and procedure; changing the inputs, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

### Thinking move · Teacher prompt

**Thinking move:** Plan

**Teacher prompt:** What is given? What is the question? Which procedure fits these facts?

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**Thinking move:** Do

**Teacher prompt:** Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

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**Thinking move:** Check

**Teacher prompt:** Does the output answer the question? Which assumption or limitation prevents a larger claim?

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Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

## **A pictorial worked case: Writing the methods section**

Methods that another team can repeat. A paper describes a toy string-classification experiment using a stated length rule.

### **The facts we will use**

Inputs: "oak", "pine", "willow". Rule: length at least 4→long, otherwise short. No whitespace trimming occurs.

## Writing the methods section

### Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



### 1. Record exact inputs

Preserve the three strings and no-trimming policy.



### 2. State procedure

Apply the inclusive length boundary at four characters.



### 3. Show resulting labels

Oak is short; pine and willow are long.

### CHANGE ONE THING

Change the rule to length greater than 4.

Explain the changed result using the matching lesson.

## Read the picture one step at a time

### Step · What happens and why

**Step:** 1. Record exact inputs

**What happens and why:** Preserve the three strings and no-trimming policy.

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**Step:** 2. State procedure

**What happens and why:** Apply the inclusive length boundary at four characters.

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### Step: 3. Show resulting labels

**What happens and why:** Oak is short; pine and willow are long.

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Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

### Why this result follows

A methods section specifies inputs, processing, boundaries, versions and scoring. Precise procedures allow a reader to repeat the computation instead of inferring it from the final labels.

### What this example does not establish

Repeating this deterministic toy rule does not validate a learned model or establish ecological categories.

### Change one thing in the pictured case

Change: Change the rule to length greater than 4.

Prediction: Pine becomes short; willow stays long and oak stays short.

Reason: The equality case distinguishes the two procedures.

### Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

## 3. Demonstrate and guide practice

### Time or stage · Teacher action

**Time or stage:** Session 1: 0–7 minutes

**Teacher action:** Retrieve the prerequisite; identify the input, procedure and evidence source.

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**Time or stage:** 7–18 minutes

**Teacher action:** Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.

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**Time or stage:** 18–32 minutes

**Teacher action:** Trace the declared procedure and worked case. Complete W2, recording intermediate steps.

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**Time or stage:** 32–45 minutes

**Teacher action:** Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.

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**Time or stage:** Session 2: 0–8 minutes

**Teacher action:** Retrieve the worked reasoning and collect the guided response before discussing it.

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**Time or stage:** 8–20 minutes

**Teacher action:** Compare a proposed boundary variant with the original case; identify what remains fixed.

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**Time or stage:** 20–35 minutes

**Teacher action:** Collect W4 independently. Require the procedure trace and the bounded conclusion.

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**Time or stage:** 35–45 minutes

**Teacher action:** Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

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Use the worked example in Sections 3 and 6. Collect a prediction before revealing its result; ask the learner to point to each intermediate fact.

Provide an input/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace. Extension: compare two concrete variants of this mechanism while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

## Reduce help deliberately

Completion task: Methods says used AI on data and got good scores; code and split are absent. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: It omits model/procedure version, data identity, units/splits, baseline, metric and scoring. Reproduction and validity cannot be assessed from the description.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

## Individual reasoning check

Use the worked case for Writing the methods section. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: A detailed methods section can reveal limitations; it does not guarantee a correct design.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

### **Pictorial practice question**

Apply the inclusive four-character rule to oak, pine and willow. Which label changes under greater-than-four, and what method detail allows another team to reproduce the distinction?

### **Reasoned teacher answer**

Preserve the three strings and no-trimming policy. Apply the inclusive length boundary at four characters. Oak is short; pine and willow are long. Changed case: Pine becomes short; willow stays long and oak stays short. The equality case distinguishes the two procedures.

Changed-case reasoning: Pine becomes short; willow stays long and oak stays short. The equality case distinguishes the two procedures.

Collect each learner’s explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

## **4. Run the investigation and diagnose misconceptions**

The worked result establishes every future case: A detailed methods section can reveal limitations; it does not guarantee a correct design.

A missing input may be replaced without recording the change: A replacement changes the evidence or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent evidence: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Section 9 for the evidence ledger. Add a boundary or missing-input case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

### **Match the support to the first error**

#### **What the response shows · Next teaching action**

**What the response shows:** Missing or misread facts

**Next teaching action:** Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

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**What the response shows:** Wrong procedure

**Next teaching action:** Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

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**What the response shows:** Arithmetic or implementation error

**Next teaching action:** Trace intermediate values or lines; preserve the first mismatch and retry that step.

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**What the response shows:** Correct result without a reason

**Next teaching action:** Ask for a trace and a changed case before marking independent understanding.

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**What the response shows:** An unsupported wider claim

**Next teaching action:** Ask which supplied evidence supports the claim and which further observation would be needed.

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Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

## 5. Assess and explain the answers

### Guided and exit questions

Guided: It omits model/procedure version, data identity, units/splits, baseline, metric and scoring. Reproduction and validity cannot be assessed from the description. Qualification: A detailed methods section can reveal limitations; it does not guarantee a correct design. Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the mechanism; reject a claim resting on unmeasured evidence.

### W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

Teacher explanation: A methods section should let a reader reconstruct the design and computation. Detail should explain choices that affect evidence. Report question, units, sampling, permissions, data roles,

baseline, procedure, metrics, uncertainty assumptions, missing/exclusion handling and versions.

Distinguish prespecified and later changes.

## **W2 Trace the worked case**

Methods says used AI on data and got good scores; code and split are absent. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: It omits model/procedure version, data identity, units/splits, baseline, metric and scoring. Reproduction and validity cannot be assessed from the description.

## **W3 Audit the limit**

A report omits this qualification: A detailed methods section can reveal limitations; it does not guarantee a correct design. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: A detailed methods section can reveal limitations; it does not guarantee a correct design. State the observed result separately from the unverified or out-of-scope claim; preserve the original evidence rather than inventing a missing fact.

## **W4 Apply a new case independently**

Methods names a dataset and metric, but not whether near-duplicate sources cross the split. What detail matters?

Teacher explanation: Describe grouping/deduplication and exact split assignment, including related examples. File names alone do not establish independence.

## **W5 Design a revealing follow-up**

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked input or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure. Require preserved settings, original outputs and data roles. A detailed methods section can reveal limitations; it does not guarantee a correct design. The new case should test this particular mechanism rather than merely restate the original result.

## **Independent assessment and fresh retry**

W4: Describe grouping/deduplication and exact split assignment, including related examples. File names alone do not establish independence. Require intermediate reasoning and the stated scope. Fresh retry: A study reports a test without its null or assumptions. Expected reasoning: Add null model, statistic, tail rule, threshold and independence/exchangeability assumptions; do not label p-values self-explanatory. If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

## Assess independent transfer

Selected case: Methods names a dataset and metric, but not whether near-duplicate sources cross the split. What detail matters?

### Selected case answer

Teacher explanation: Describe grouping/deduplication and exact split assignment, including related examples. File names alone do not establish independence.

#### Dimension · Evidence required

**Dimension:** Explanation

**Evidence required:** Names the applicable idea or procedure and ties it to the supplied facts.

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**Dimension:** Trace and result

**Evidence required:** Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

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**Dimension:** Limits

**Evidence required:** States the assumptions, missing evidence or conditions under which the result would change.

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Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

### Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

## 6. Connect the project, support and next lesson

Add a labelled writing the methods section evidence sheet to Reproducible AI Research Study. Show how you can explain the mechanism and terminology of writing the methods section. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an input/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace. Extension: compare two concrete variants of this mechanism while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Writing results and limitations. Preserve the current case record and retrieve its evidence boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

## Plan the delayed return

Suggested return points in this level: ai-7/31, ai-7/33, ai-7/36. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

### Record · Teacher evidence

**Record:** Learner code and date

**Teacher evidence:** \_\_\_\_\_  
\_\_\_\_\_

**Record:** Concept and case used; prior exposure

**Teacher evidence:** \_\_\_\_\_  
\_\_\_\_\_

**Record:** Explanation / trace / limits

**Teacher evidence:** \_\_\_\_\_  
\_\_\_\_\_

**Record:** Support and first repair

**Teacher evidence:** \_\_\_\_\_  
\_\_\_\_\_

**Record:** Changed case and delayed outcome

**Teacher evidence:** \_\_\_\_\_  
\_\_\_\_\_

**Record:** Next teaching action

**Teacher evidence:** \_\_\_\_\_

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

## Project evidence from this case

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

## 7. Prepare, teach and assess

Teaching target: Explain the mechanism and terminology of writing the methods section

Prepare the labelled worked-case picture, the matching workbook, fictional input cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

### Session · Plan · Resources

**Session:** Session 1 · Understand · 45 min

**Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

**Resources:** Sections 1–6; record the brief recall in the workbook.

**Session:** Session 2 · Practise · 45 min

**Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

**Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

**Session:** Session 3 · Investigate and transfer · 45 min

**Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

**Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: Methods says used AI on data and got good scores; code and

split are absent. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: It omits model/procedure version, data identity, units/splits, baseline, metric and scoring. Reproduction and validity cannot be assessed from the description.

Check every learner before discussion: Use the worked case for Writing the methods section. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: A detailed methods section can reveal limitations; it does not guarantee a correct design.

Support: read the exact case aloud, enlarge the labelled picture, supply input cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Change the rule to length greater than 4. Require a prediction and a reason before comparing. Expected result: Pine becomes short; willow stays long and oak stays short. Reason: The equality case distinguishes the two procedures.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

### **Criterion · Not yet · With support · Independent**

**Criterion:** Explanation

**Not yet:** Names an answer without a deciding fact

**With support:** Uses a hint to link fact and decision

**Independent:** Explains the deciding fact in own words

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**Criterion:** Trace or test

**Not yet:** Cannot connect input and result

**With support:** Completes steps with prompts

**Independent:** Traces or tests a changed case accurately

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**Criterion:** Limits and revision

**Not yet:** Claims more than evidence supports

**With support:** Names a limit after a prompt

**Independent:** States a defensible limit and useful next test

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Project checkpoint: Prepare the demonstration. Gather reproducible instructions, evidence, access checks

and remaining limitations.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and evidence.